

# SatsPath × Arkade — Technical Integration Plan (Production / v2)

**Goal:** Fully native, self-sovereign SatsPath stack in Arkade wallet (mobile + desktop) with zero trusted backend. UniFFI bindings for Kotlin (Android), Swift (iOS), TypeScript (React Native / Tauri / PWA).

## Architecture:

```
Arkade Wallet (Kotlin/Swift/TS)
  ↓ UniFFI
satspath-core + satspath-router (Rust cdylib)
  Resolver Chain (local → BIP353 → HTTPS → Nostr → P2P)
  Profile Crypto (secp256k1 sign/verify)
  Router (fee-aware rail selection)
  QR/Payment Payload Builder
  ↓ Execution Adapters (platform-specific)
LDK (Lightning) / Arkade SDK (Ark) / BDK (On-chain)
```

## 1. Production Requirements (vs MVP)

| Area                                                                                | MVP                                        | Production                                                                                                                                                             |
|-------------------------------------------------------------------------------------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Resolver Crypto</b>                                                              | TS fetch in PWA<br>WASM (secp256k1 only)   | Rust async in core (tokio) — no JS fetch dependency<br>Native Rust (secp256k1 + schnorrkel) via UniFFI                                                                 |
| <b>Router Identity</b>                                                              | TS port (~300 LOC)<br>LocalStorage profile | Native Rust (full scoring, urgency, split payments)<br>Encrypted SQLCipher/Keychain/Keystore + BIP-39 seed backup                                                      |
| <b>P2P</b>                                                                          | Optional satspathd sidecar                 | Embedded Hyperswarm via UniFFI (or libp2p)                                                                                                                             |
| <b>Ark BOLT12 Silent Payments Split Payments Key Rotation Platform Verification</b> | Testnet preview<br>Invite only             | Mainnet ready (covenant-based when available)<br>Offer-based async payments<br>BIP-352 receive<br>Multi-rail single payment<br>Spec §29<br>DKIM / OAuth / Nostr NIP-42 |

## 2. Rust Refactor for UniFFI

### 2.1 Crate Restructuring

```
satspath/
  crates/
    satspath-core/      # → Keep, make `cdylib` + `uniffi` feature
    satspath-router/    # → Keep, make `cdylib` + `uniffi` feature
    satspath-ffi/       # NEW: UniFFI definitions (UDL) + glue
      satspath.udl
      build.rs
      src/
        resolver.rs     # ResolverChain implementation
        router.rs       # Router FFI wrapper
```

```

        profile.rs      # Profile management FFI
        crypto.rs       # Crypto FFI wrapper
    satspath-wasm/      # Keep for web-only fallback

```

## 2.2 satspath-core Changes

Cargo.toml:

```

[lib]
crate-type = ["rlib", "cdylib"] # cdylib for UniFFI

[features]
default = []
uniffi = ["uniffi", "tokio/rt-multi-thread", "request/native-tls"]

[dependencies]
uniffi = { version = "0.28", optional = true }
tokio = { version = "1", features = ["rt-multi-thread", "macros"], optional = true }
request = { version = "0.12", features = ["json", "native-tls"], optional = true }
# ... existing deps

```

Expose async trait for UniFFI (src/resolver.rs):

```

use uniffi::async_trait;

#[async_trait]
pub trait ProfileResolver: Send + Sync {
    async fn resolve_alias(&self, alias: &str) -> Result<SignedPaymentProfile, SatsPathError>;
}

// Implement for each resolver: LocalRegistry, Bip353, HttpWellKnown, NostrNip05, P2P

```

## 2.3 UDL Definitions (satspath-ffi/satspath.udl)

```

namespace satspath;

interface ProfileResolver {
    Promise<SignedPaymentProfile> resolveAlias(string alias);
}

enum PaymentMethodType { Onchain, Lightning, Ark };

record OnchainMethod {
    string label;
    string network; // "mainnet" | "testnet" | "regtest"
    string? address;
    string? silentPaymentPubkey;
    string? pubkeyHint;
    string? descriptorHint;
    sequence<string> addressList;
};

record LightningMethod {
    string label;
    string? lightningAddress;
    string? lnurl;
}

```

```

    string? bolt12;
    string? receiverPubkey;
};

record ArkMethod {
    string label;
    string server;
    string pubkey;
    string? vtxoPointer;
    string? opaqueUri;
    ArkOwnershipProof? proof;
    i64? expiresAt;
};

union PaymentMethod { OnchainMethod | LightningMethod | ArkMethod };

record PaymentProfile {
    string alias;
    string identityPubkey;
    sequence<PaymentMethod> methods;
    i64 updatedAt;
    i64? expiresAt;
    u64? sequence;
    sequence<string> preferences;
    string? nonce;
    KeyRotation? rotation;
    sequence<MethodVerification> methodVerifications;
};

record SignedPaymentProfile {
    PaymentProfile profile;
    string signature; // hex schnorr
};

record QuoteRequest {
    string recipient;
    u64 amountSats;
    SignedPaymentProfile signedProfile;
    string urgency; // "low" | "normal" | "high"
    u64? maxFeeSats;
    f64? maxFeePercent;
};

enum ExecutionMode { Preview, MainnetPreview, TestnetExperimental, ManualWallet };

record RouteQuote {
    PaymentMethod selectedMethod;
    u64 estimatedFeeSats;
    string estimatedConfirmation;
    string reason;
    ExecutionMode execution;
    string walletHint;
};

```

```

record FeeEstimate {
    u64 fastestFee;
    u64 halfHourFee;
    u64 hourFee;
    u64 economyFee;
    u64 minimumFee;
};

interface Router {
    Promise<RouteQuote> selectRoute(QuoteRequest request, FeeEstimate fees);
    Promise<FeeEstimate> fetchFeeEstimate();
    string buildQrPayload(PaymentMethod method, u64 amountSats);
};

enum QuoteStatus { Ok, NotRegistered, NoRoute, InvalidSignature };

record QuoteRecipient {
    string alias;
    boolean verified;
    boolean profileSignatureVerified;
    boolean identifierVerified;
    string identifierVerification;
    string fingerprint;
};

record Invite {
    string aliasHash;
    u64 amountSats;
    i64 createdAt;
    i64 expiresAt;
    string claimUrl;
    string warning;
    string? senderSignature;
    string? senderPubkey;
};

union QuoteResponse {
    record { QuoteRecipient recipient; PaymentMethod selectedMethod; u64 feeSats; string eta; string reason; }
    record { Invite invite; },
    record { string reason; },
    record { QuoteRecipient recipient; }
};

interface Satspath {
    // High-level: resolve + verify + route + build payload
    Promise<QuoteResponse> quote(string recipient, u64 amountSats);

    // Lower-level building blocks
    Promise<SignedPaymentProfile> resolve(string alias);
    boolean verifyProfile(SignedPaymentProfile profile);
    Promise<RouteQuote> route(QuoteRequest request);

    // Identity management
    Promise<Identity> createIdentity();
};

```

```

    Promise<void> saveProfile(SignedPaymentProfile profile);
    Promise<SignedPaymentProfile?> loadProfile(string alias);

    // P2P (optional)
    Promise<void> startP2pBridge(string profilePath);
    void stopP2pBridge();
};

record Identity {
    string pubkeyHex;
    string secretKeyPath; // encrypted, platform keystore
    string fingerprint;
};

```

## 2.4 FFI Implementation (satspath-ffi/src/)

```

// satspath-ffi/src/resolver.rs
use uniffi::async_trait;
use satspath_core::resolver::{ChainResolver, ProfileResolver as CoreResolver};
use satspath_core::resolvers::{bip353::Bip353Resolver, http::HttpResolver, nostr::NostrResolver};

pub struct ResolverChain {
    inner: ChainResolver,
}

#[uniffi::export(async_runtime = "tokio")]
impl ResolverChain {
    #[uniffi::constructor]
    pub fn new() -> Self {
        let mut chain = ChainResolver::new();
        chain = chain.push(LocalRegistryResolver::new()); // impl CoreResolver
        chain = chain.push(Bip353Resolver::new());
        chain = chain.push(HttpResolver::new());
        chain = chain.push(NostrResolver::new());
        // P2P optional
        Self { inner: chain }
    }

    pub async fn resolve_alias(&self, alias: String) -> Result<SignedPaymentProfile, SatsPathError> {
        self.inner.resolve_alias(&alias).await
    }
}

// satspath-ffi/src/router.rs
use satspath_router::{select_route, select_route_with_fees, FeeEstimate, RouteRequest, build_qr_payload};

#[uniffi::export(async_runtime = "tokio")]
pub async fn select_route_ffi(request: QuoteRequest, fees: FeeEstimate) -> Result<RouteQuote, SatsPathError> {
    let req = RouteRequest {
        alias: request.recipient,
        amount_sats: request.amount_sats,
        signed_profile: request.signed_profile.into(),
        urgency: parse_urgency(request.urgency),
        max_fee_sats: request.max_fee_sats,
    };

```

```

        max_fee_percent: request.max_fee_percent,
    };
    let quote = select_route_with_fees(&req, &fees)?;
    Ok(quote.into())
}

#[uniffi::export(async_runtime = "tokio")]
pub async fn fetch_fee_estimate_ffi() -> Result<FeeEstimate, SatsPathError> {
    satspath_router::fees::fetch_fee_estimate().await.map_err(Into::into)
}

// satspath-ffi/src/profile.rs
#[uniffi::export]
pub fn verify_profile(profile: SignedPaymentProfile) -> bool {
    satspath_core::crypto::verify_signed_profile(&profile.into()).unwrap_or(false)
}

#[uniffi::export(async_runtime = "tokio")]
pub async fn quote(recipient: String, amount_sats: u64) -> QuoteResponse {
    let resolver = ResolverChain::new();
    satspath_router::quote_with_resolver(&resolver, &recipient, amount_sats).await.into()
}

```

## 2.5 Build Pipeline

```

# 1. Generate bindings
cd satspath-ffi
cargo build --release --features uniffi
cargo run --features uniffi --bin uniffi-bindgen generate \
    --library target/release/libsatpath_ffi.so \
    --language kotlin --out-dir ../../arkade-wallet/android/satspath/src/main/java
cargo run --features uniffi --bin uniffi-bindgen generate \
    --library target/release/libsatpath_ffi.so \
    --language swift --out-dir ../../arkade-wallet/ios/Satspath
cargo run --features uniffi --bin uniffi-bindgen generate \
    --library target/release/libsatpath_ffi.so \
    --language typescript --out-dir ../../arkade-wallet/src/satspath

# 2. Android: copy .so to jniLibs
# 3. iOS: embed .framework in Xcode
# 4. TS: npm install @satspath/core (or file:../satspath)

```

---

## 3. Platform-Specific Integration

### 3.1 Android (Kotlin + LDK + Arkade SDK + BDK)

Gradle (android/app/build.gradle.kts):

```

dependencies {
    implementation("org.ldk:ldk-android:0.0.120") // or Breez SDK
    implementation("org.bitcoindevkit:bdk-android:1.0.0")
    implementation(project(":satspath")) // UniFFI generated module
    // Arkade SDK as local module or maven
}

```

Identity Storage (android/satspath/src/main/java/com/satspath/IdentityManager.kt):

```
class IdentityManager @Inject constructor(
    @Assisted private val context: Context
) {
    private val keystore = KeyStore.getInstance("AndroidKeyStore")
    private val prefs = context.getSharedPreferences("satspath", Context.MODE_PRIVATE)

    fun createIdentity(): Identity {
        val kp = Satspath.createIdentity() // UniFFI call
        // Store secret in Keystore, pubkey in SharedPreferences
        storeSecret(kp.secretKeyPath)
        prefs.edit().putString("pubkey", kp.pubkeyHex).apply()
        return kp
    }

    fun signProfile(profile: PaymentProfile): SignedPaymentProfile {
        val secret = loadSecret() // from Keystore
        return Satspath.signProfile(profile, secret)
    }
}
```

Send Flow (android/ui/send/SendViewModel.kt):

```
class SendViewModel @ViewModelInject constructor(
    private val satspath: Satspath,
    private val ldk: LdkManager,
    private val arkade: ArkadeSdk,
    private val bdk: BdkManager
) : ViewModel() {

    fun getQuote(alias: String, amountSats: Long) = viewModelScope.launch {
        val response = satspath.quote(alias, amountSats)
        _quoteState.value = response
    }

    fun executePayment(response: QuoteResponse) = viewModelScope.launch {
        when (response.status) {
            QuoteStatus.Ok -> {
                val method = response.selectedMethod
                when (method) {
                    is PaymentMethod.LightningMethod -> {
                        val invoice = if (response.qr.startsWith("lnbc")) response.qr
                        else ldk.fetchInvoice(response.qr, response.feeSats)
                        ldk.sendPayment(invoice)
                    }
                    is PaymentMethod.OnchainMethod -> {
                        bdk.sendPayment(response.qr) // BIP-21 URI
                    }
                    is PaymentMethod.ArkMethod -> {
                        arkade.sendPayment(response.qr) // ark: URI
                    }
                }
            }
            QuoteStatus.NotRegistered -> showInviteFlow(response.invite)
        }
    }
}
```

```

    }
}

```

### 3.2 iOS (Swift + LDK + Arkade SDK + BDK)

Package.swift:

```

dependencies: [
    .package(url: "https://github.com/lightningdevkit/ldk-swift", from: "0.1.0"),
    .package(url: "https://github.com/bitcoinddevkit/bdk-swift", from: "1.0.0"),
    // Satspath framework embedded in Xcode
]

```

Identity (SatspathIdentityManager.swift):

```

class SatspathIdentityManager {
    let keychain = Keychain(service: "com.arkade.satspath")

    func createIdentity() throws -> Identity {
        let identity = try Satspath.createIdentity() // UniFFI
        try keychain.set(identity.secretKeyPath, key: "satspath_secret_\(identity.fingerprint)")
        return identity
    }

    func signProfile(_ profile: PaymentProfile) throws -> SignedPaymentProfile {
        let secret = try keychain.get("satspath_secret_\(currentFingerprint)")
        return try Satspath.signProfile(profile, secret)
    }
}

```

Send Flow (SendViewModel.swift):

```

@MainActor
class SendViewModel: ObservableObject {
    @Published var quote: QuoteResponse?

    func getQuote(alias: String, amount: UInt64) async {
        quote = try await Satspath.shared.quote(alias, amount)
    }

    func pay() async throws {
        guard case .ok(let data) = quote?.status else { return }
        switch data.selectedMethod {
        case .lightning(let method):
            let invoice = data.qr.hasPrefix("lnbc") ? data.qr : try await LDKManager.shared.fetchInvoice
            try await LDKManager.shared.sendPayment(invoice)
        case .onchain(let method):
            try await BDKManager.shared.sendPayment(data.qr)
        case .ark(let method):
            try await ArkadeSDK.shared.sendPayment(data.qr)
        }
    }
}

```



### 3.3 React Native / Tauri / PWA (TypeScript)

npm dependency: @satspath/core (generated from UniFFI TS bindings)

```
// packages/satspath-core/src/index.ts (generated)
export interface Satspath {
  quote(recipient: string, amountSats: bigint): Promise<QuoteResponse>;
  resolve(alias: string): Promise<SignedPaymentProfile>;
  verifyProfile(profile: SignedPaymentProfile): boolean;
  createIdentity(): Promise<Identity>;
  // ...
}

export const satspath: Satspath; // WASM fallback if native not available

Usage (same as MVP but native):

import { satspath } from "@satspath/core";

const response = await satspath.quote("alice@example.com", 21000n);
if (response.status === "ok") {
  switch (response.selected_method.type) {
    case "Lightning":
      await ldkSend(response.qr);
      break;
    case "Onchain":
      await bdkSend(response.qr);
      break;
    case "Ark":
      await arkadeSend(response.qr);
      break;
  }
}
```

---

## 4. Ark Mainnet Readiness (Production Blockers)

| Blocker                     | Status                     | Solution                                                           |
|-----------------------------|----------------------------|--------------------------------------------------------------------|
| Covenant opcodes (CTV/CSFS) | Not on mainnet             | Use connector outputs + pre-signed txs (current Ark design)        |
| Ark server trust            | ASP can steal if malicious | Client-side VTXO verification (Arkade SDK Tier 1-3) — already done |
| Unilateral exit UX          | Complex                    | Arkade SDK sovereignStorage + auto-broadcast on expiry             |
| Liquidity                   | Limited                    | Boltz submarine swaps for on-ramp/off-ramp                         |
| Fee estimation              | Ark server sets            | Router queries ASP /fee endpoint, falls back to mempool            |

Router Ark Logic (enhance ark\_routes.rs):

```
pub async fn plan_ark_route_mainnet(
  sender: &SenderCapabilities,
```

```

    receiver: &SignedPaymentProfile,
    asp_fees: &AspFeeInfo,
) -> Option<ArkRoutePlan> {
    // Verify ASP fee policy is reasonable
    // Check receiver VTXO expiry > 144 blocks (safety margin)
    // Ensure sender has VTXOs on same ASP or can swap via Boltz
    // Return ArkToArk / ArkToLightning / ArkToOnchain with real fee estimates
}

```

---

## 5. BOLT12 / Silent Payments / Split Payments (v2.1+)

### 5.1 BOLT12 Offers

```

// satspath-core/src/profile.rs - extend PaymentMethod::Lightning
Lightning {
    // ... existing
    bolt12: Option<String>, // offer: lno1...
    // Add: offer fetching + invoice_request flow
}

```

Router: if bolt12 present → fetch offer → send invoice\_request → pay invoice.

### 5.2 Silent Payments (BIP-352)

```

Onchain {
    silent_payment_pubkey: Option<String>, // sp1q...
    // Payer derives: address = hash(sp_pubkey + shared_secret)
}

```

Router: if silent\_payment\_pubkey present → derive unique address per payment.

### 5.3 Split Payments

```

// New route type
enum RouteQuote {
    Single(RouteQuote),
    Split(Vec<SplitRoute>), // { method, amount_sats, fee_sats } summing to total
}

```

Scoring: minimize total fee, respect per-rail limits.

---

## 6. Key Rotation & Profile Revocation (Spec §29)

UDL Addition:

```

record KeyRotation {
    string newIdentityPubkey;
    i64 rotationTime;
    string previousSignature; // sig by old key over new key
};

interface Satspath {
    Promise<SignedPaymentProfile> rotateKey(string alias, string newPubkeyHex);
}

```

```
    boolean verifyRotation(SignedPaymentProfile oldProfile, SignedPaymentProfile newProfile);
}
```

**Flow:** 1. User generates new identity key (in Secure Enclave/Keystore) 2. Sign rotation object with old key 3. Publish new profile with rotation field + both signatures 4. Resolvers verify chain: old\_sig verifies rotation + new\_sig verifies profile

---

## 7. Platform Verification (Email/Domain Ownership)

**Current:** Invite flow only (no proof)

**Production:** | Method | Verification | UDL | |——|———|——| | **DKIM** | User forwards verification email → wallet verifies DKIM sig | **EmailVerifier** interface | | **Nostr NIP-42** | Relay challenges user to sign event | **NostrVerifier** | | **DNS** | User adds TXT record → resolver checks | **DnsVerifier** (BIP-353) | | **OAuth** | Wallet opens auth flow → gets signed attestation | **OAuthVerifier** |

**Profile Field:**

```
record MethodVerification {
    string methodDescriptor; // PaymentMethod::ownership_descriptor()
    string proofType;        // "dkim" | "nostr" | "dns" | "oauth"
    string proofData;        // JSON attestation
    i64 verifiedAt;
}
```

---

## 8. P2P Transport (Optional but Sovereign)

**Current:** satspathd spawns Node.js Holepunch bridge

**Production:** Embed Hyperswarm in Rust via hyperdriver (or libp2p)

```
// satspath-core/src/p2p.rs
pub struct P2pResolver {
    swarm: HyperSwarm,
    topic: Topic,
}

#[async_trait]
impl ProfileResolver for P2pResolver {
    async fn resolve_alias(&self, alias: &str) -> Result<SignedPaymentProfile, SatsPathError> {
        let topic = derive_topic(alias); // SHA256("satspath-profile:" + alias)
        let mut stream = self.swarm.join(topic).await?;
        // Request profile via DHT
        let profile = stream.request(alias).await?;
        verify_signed_profile(&profile)?;
        Ok(profile)
    }
}
```

**UniFFI:** Expose startP2p() / stopP2p() — mobile runs swarm in background.

## 9. Testing & CI/CD (Production)

### 9.1 Rust Layer

```
# .github/workflows/rust.yml
- cargo test --workspace --features uniffi
- cargo clippy --workspace --features uniffi -- -D warnings
- cargo fmt --check
- cargo audit
```

### 9.2 UniFFI Bindings Tests

```
# Kotlin
cd android && ./gradlew test

# Swift
cd ios && xcodebuild test -scheme SatspathTests

# TypeScript
cd sdk/ts && npm test
```

### 9.3 Integration Tests (Testnet)

```
// tests/integration/mainnet_preview.rs
#[tokio::test]
async fn mainnet_preview_quote() {
    let satspath = Satspath::new();
    let quote = satspath.quote("rodrigo@satspath.dev", 100_000).await;
    assert!(matches!(quote.status, QuoteStatus::Ok));
    assert!(quote.qr.starts_with("bitcoin:") || quote.qr.starts_with("lnbc"));
}
```

### 9.4 Fuzz Testing

```
cargo fuzz run profile_parsing
cargo fuzz run router_scoring
```

---

## 10. File Tree (Production Additions)

```
satspath/
├── crates/
│   ├── satspath-core/
│   │   ├── Cargo.toml          # +uniffi feature, cdylib
│   │   └── src/
│   │       ├── p2p.rs          # NEW: embedded Hyperswarm
│   │       ├── rotation.rs     # ENHANCE: key rotation logic
│   │       └── verification.rs # NEW: DKIM/Nostr/DNS/OAuth verifiers
│   ├── satspath-router/
│   │   └── src/
│   │       ├── bolt12.rs       # NEW: offer → invoice_request flow
│   │       ├── silent.rs       # NEW: BIP-352 address derivation
│   │       └── split.rs        # ENHANCE: multi-rail routing
│   ├── satspath-ffi/
│   │   └── Cargo.toml         # NEW: UniFFI layer
│   └── ...
```

```

    build.rs
    satspath.udl
    src/
        lib.rs
        resolver.rs
        router.rs
        profile.rs
        identity.rs
        p2p.rs
    satspath-wasm/          # Keep for web fallback
sdk/
    kotlin/                # Generated by UniFFI
    swift/                  # Generated by UniFFI
    typescript/             # Generated by UniFFI
arkade-wallet/
    android/
        satspath/          # UniFFI Kotlin module
    ios/
        Satspath/          # UniFFI Swift module
    src/
        satspath/          # UniFFI TS module
docs/
    TECHNICAL_PLAN_MVP.md
    TECHNICAL_PLAN_PRODUCTION.md

```

## 11. Timeline (Production)

| Phase                              | Weeks     | Deliverable                                                                          |
|------------------------------------|-----------|--------------------------------------------------------------------------------------|
| <b>0. Prep</b>                     | 1         | Enable <code>uniffi</code> feature, <code>cdylib</code> , fix WASM-incompatible deps |
| <b>1. FFI Layer</b>                | 3         | <code>satspath-ffi</code> crate + UDL + all bindings (Kotlin/Swift/TS)               |
| <b>2. Resolver Chain</b>           | 2         | Port all resolvers to async Rust trait, remove TS resolvers                          |
| <b>3. Router Native</b>            | 2         | Full router in Rust (scoring, urgency, split, BOLT12, silent)                        |
| <b>4. Identity &amp; Storage</b>   | 2         | Encrypted keystore (Keystore/Keychain/SQLCipher) + BIP-39 backup                     |
| <b>5. P2P Embedded</b>             | 3         | <code>hyperdriver</code> integration, background swarm on mobile                     |
| <b>6. Platform Verification</b>    | 2         | DKIM + NIP-42 + DNS verifiers                                                        |
| <b>7. Ark Mainnet</b>              | 3         | Covenant-ready ASP integration, fee estimation, liquidity                            |
| <b>8. Mobile Integration</b>       | 4         | Arkade Android/iOS wiring (LDK, BDK, Arkade SDK)                                     |
| <b>9. Auditing &amp; Hardening</b> | 3         | Security audit, fuzzing, testnet soak, mainnet preview                               |
| <b>Total</b>                       | ~25 weeks | <b>Production-ready Arkade + SatsPath</b>                                            |

## 12. Definition of Done (Production)

- ☐ `cargo build --release --features uniffi` produces `.so/.dylib/.dll` + bindings for 3 targets
  - ☐ Android app: `satspath.quote("alice@example.com", 100_000)` returns in <500ms (no network on cached profile)
  - ☐ iOS app: same, using Swift UniFFI module
  - ☐ React Native/Tauri: same, using TS bindings
  - ☐ Key rotation: user rotates key → new profile verifies on all resolvers
  - ☐ P2P: profile resolves via Hyperswarm without HTTPS/DNS
  - ☐ Ark mainnet: send/receive VTXO on mainnet ASP, sovereign exit tested
  - ☐ BOLT12: pay offer → invoice\_request → invoice → pay
  - ☐ Silent Payments: derive unique address per payment from `sp1q...`
  - ☐ Split Payments: single 500k sat payment splits 300k LN + 200k on-chain
  - ☐ DKIM verification: user proves `alice@gmail.com` ownership
  - ☐ All integration tests pass on testnet (signet + mutiny.net)
  - ☐ Security audit completed (crypto, memory safety, supply chain)
- 

## 13. Risk Register (Production)

| Risk                             | Likelihood | Impact | Mitigation                                                                       |
|----------------------------------|------------|--------|----------------------------------------------------------------------------------|
| UniFFI async trait limitations   | Medium     | High   | Use <code>uniffi::async_trait</code> + <code>tokio</code> runtime; test early    |
| Mobile binary size (Rust + deps) | High       | Medium | <code>strip</code> , <code>lto</code> , feature-gate optional modules (P2P, Ark) |
| Ark mainnet covenant delay       | High       | High   | Ship with connector-output design; upgrade when CTV activates                    |
| Platform keystore differences    | Medium     | Medium | Abstract <code>SecureStorage</code> trait per platform                           |
| Nostr relay censorship           | Low        | Medium | Multi-relay fallback; P2P as ultimate fallback                                   |
| Regulatory (KYC/AML on identity) | Low        | High   | SatsPath = identity only, no custody; wallet handles compliance                  |